Supporting Information

Synergistic Buried Interface Regulation of Tin-Lead Perovskite Solar Cells via Co-Self-Assembled Monolayers

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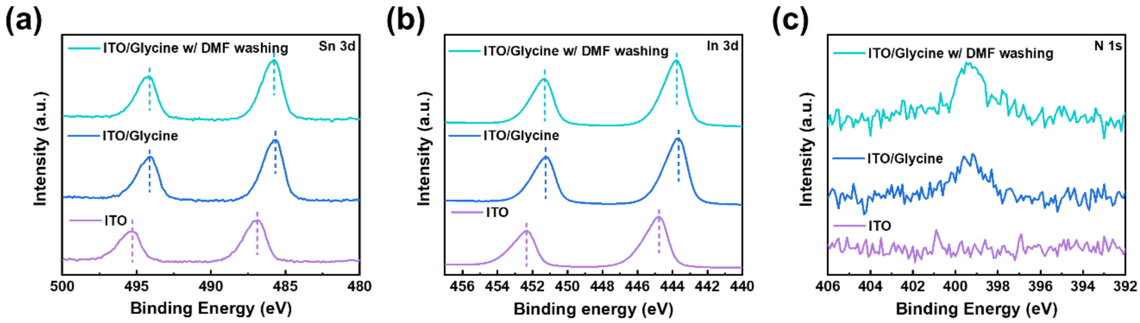

Figure S1. XPS spectra of (a) Sn 3d, (b) In 3d, and (c) N 1s of ITO (purple), glycine-modified substrate before (blue) and after (green) DMF washing.

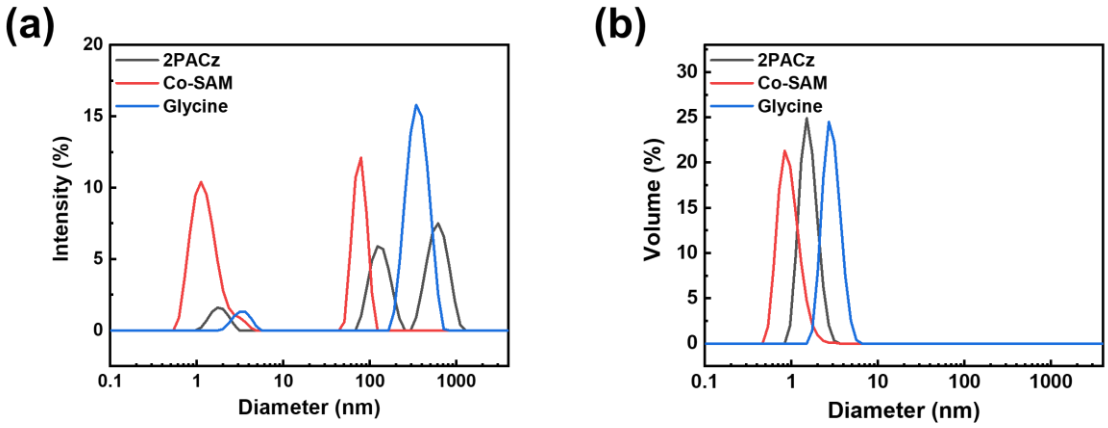

Figure S2. DLS particle size distributions in SAM-processing solutions. (a) Intensity-weighted and (b) volume-weighted distributions of 1mM 2PACz (grey), Co-SAM (red) (2PACz:glycine in a molar ratio of 5:5), and glycine (blue) processing solutions dissolved in anhydrous DMF.

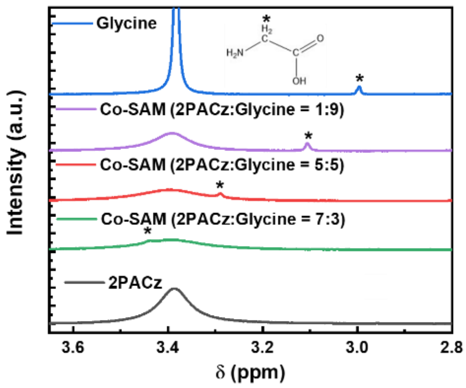

Figure S3. ^{1}H NMR spectra of 2PACz, Co-SAM, and glycine dissolved in DMSO-d_6. The molar ratios of 2PACz:glycine in the Co-SAM solutions are 1:9, 5:5, and 7:3.

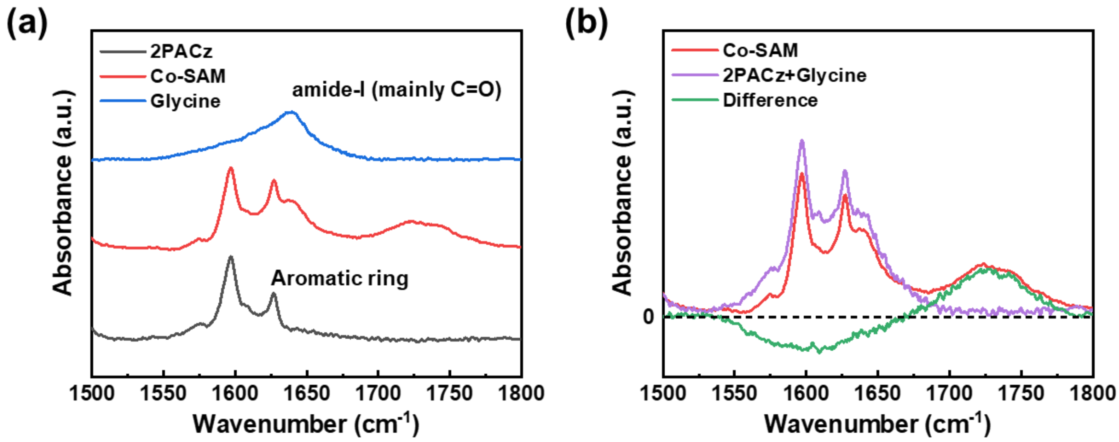

Figure S4. FTIR spectra of 2PACz, Co-SAM, and glycine dissolved in DMSO. The molar ratio of 2PACz:glycine in the Co-SAM solution is 5:5.

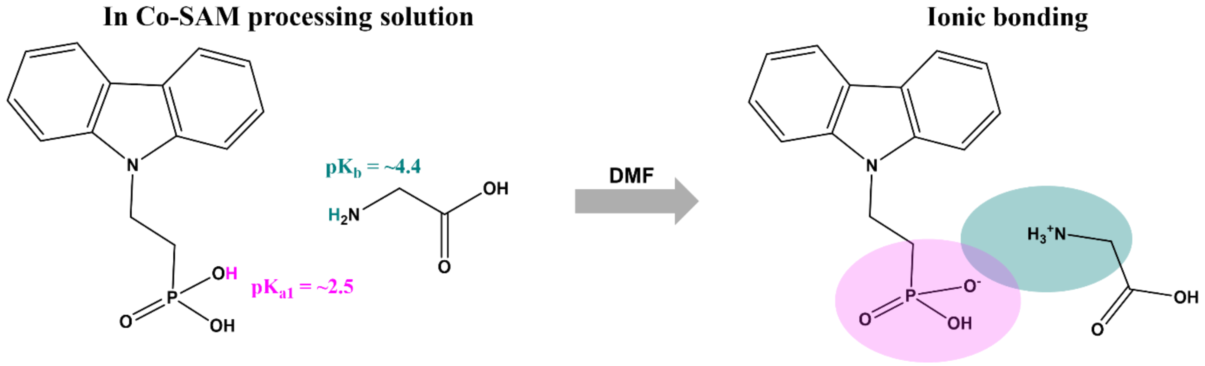

Figure S5. Molecular interactions between 2PACz and glycine dissolved in the Co-SAM processing solution.

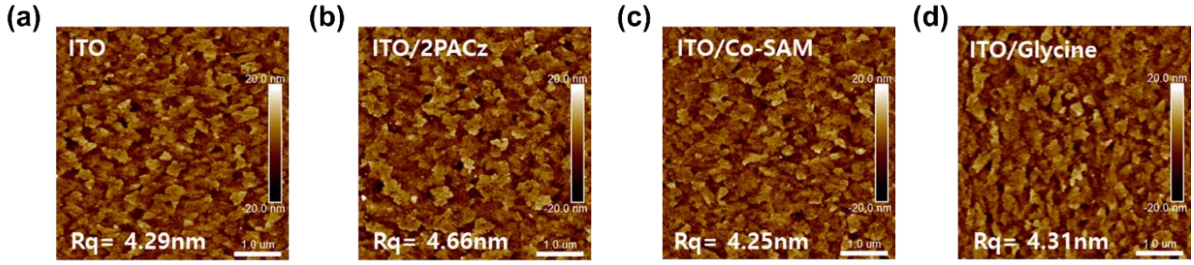

Figure S6. AFM images of (a) ITO, (b) ITO/2PACz, (c) ITO/Co-SAM, and (d) ITO/glycine.

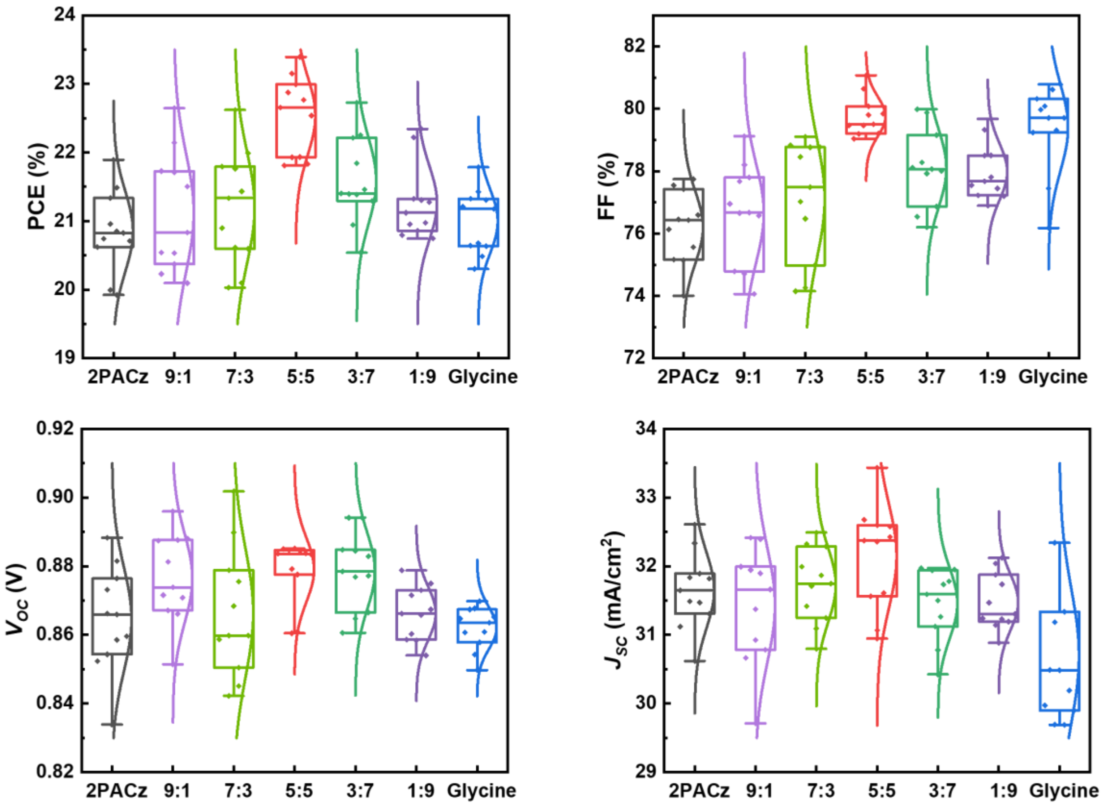

Figure S7. Distributions of the photovoltaic parameters PCE, FF, V_{OC}, and J_{SC} of Sn-Pb PSCs with Co-SAM deposited by various ratios of 2PACz to glycine.

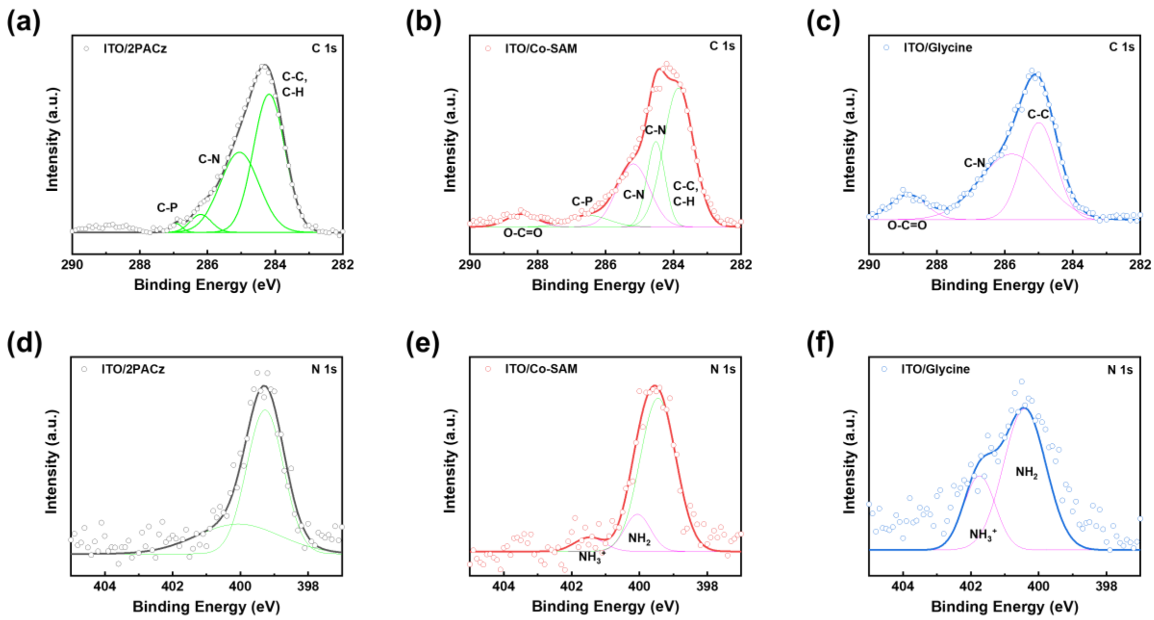

Figure S8. (a-c) C 1s and (d-f) N 1s XPS spectra of (a,d) 2PACz-, (b,e) Co-SAM-, and (c,f) glycine-modified ITO substrates. The grey, red, blue solid lines show the fit to the data and the green and pink peaks show the components thereof. The green peak fits the N from carbazole.

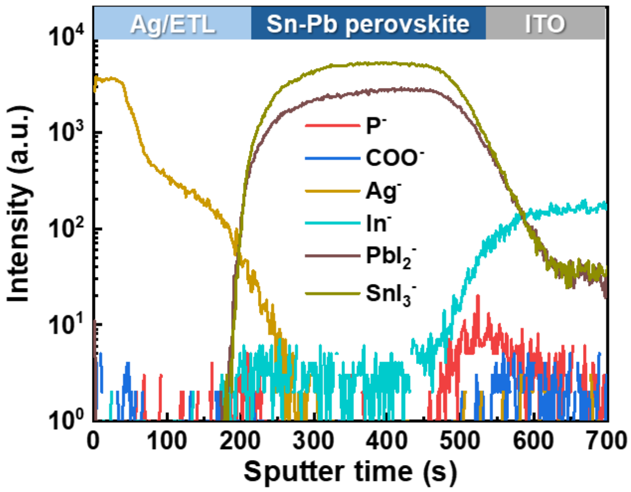

Figure S9. ToF-SIMS depth profiles of P^-, COO^-, Ag^-, In^-, PbI_2^-, and SnI_3^- ions in device using Co-SAM, after being subjected to continuous light exposure and bias for 2 hours without encapsulation (40 ℃, 60% RH).

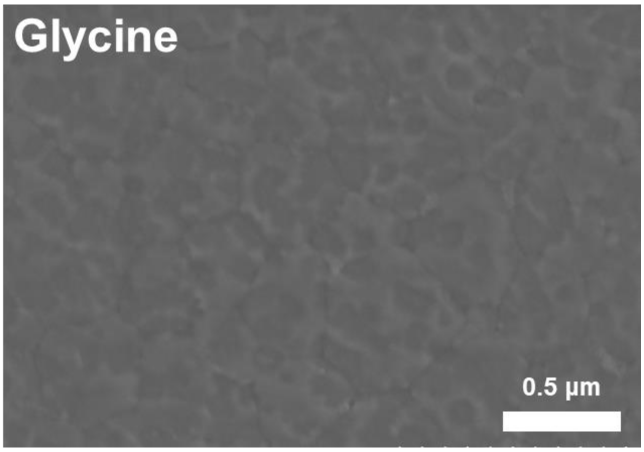

Figure S10. Buried interface SEM image of a perovskite film coated on ITO/glycine.

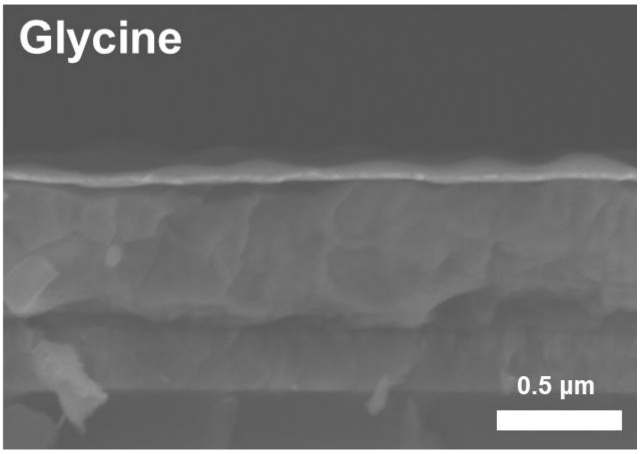

Figure S11. Cross-sectional SEM image of a perovskite film coated on ITO/glycine.

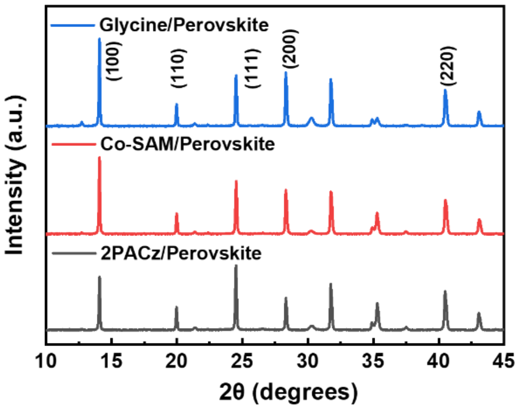

Figure S12. XRD patterns of perovskite films coated on ITO/2PACz, ITO/Co-SAM, and ITO/glycine.

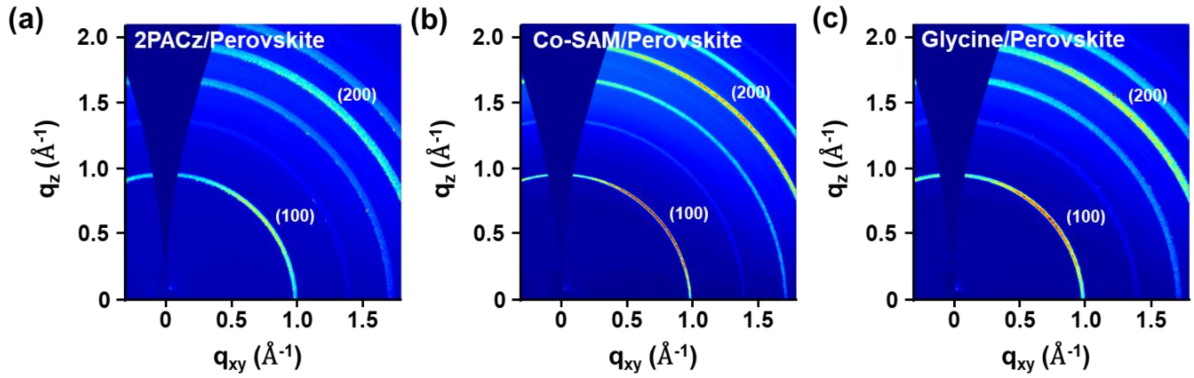

Figure S13. GIWAXS images of the perovskite films deposited on the (a) ITO/2PACz, (b) ITO/Co-SAM, and (c) ITO/glycine substrates.

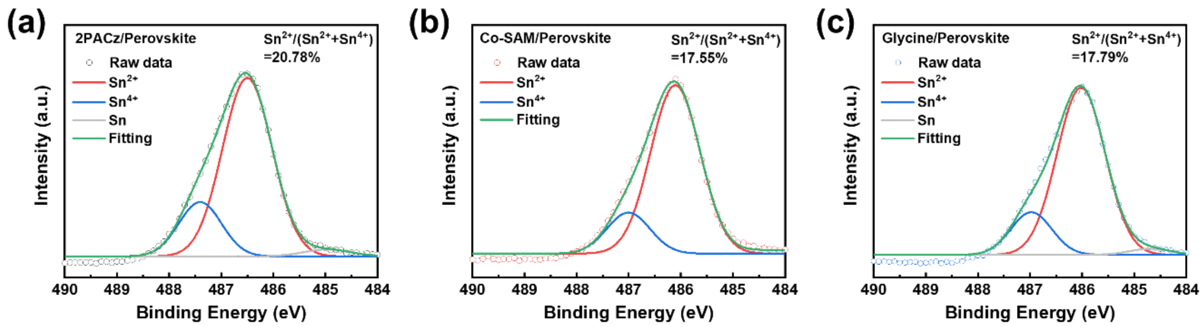

Figure S14. Sn $3d_{5/2}$ XPS spectra of perovskite films coated on (a) ITO/2PACz, (b) ITO/Co-SAM, and (c) ITO/glycine.

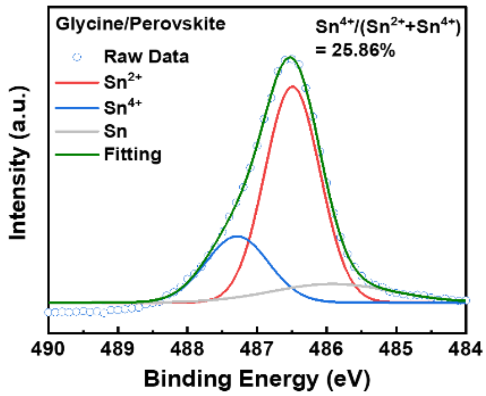

Figure S15. Sn $3d_{5/2}$ XPS spectra of perovskite film coated on ITO/glycine after aging at 65 ℃ for 40 hours.

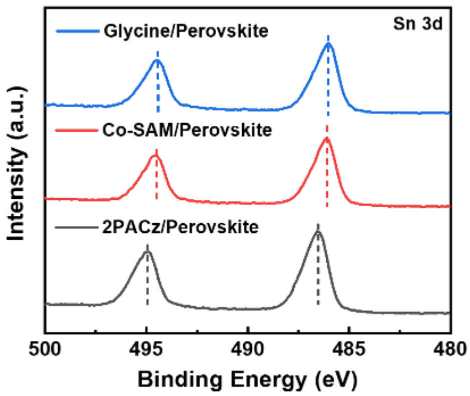

Figure S16. Sn 3d XPS spectra of perovskite films coated on ITO/2PACz, ITO/Co-SAM, and ITO/glycine.

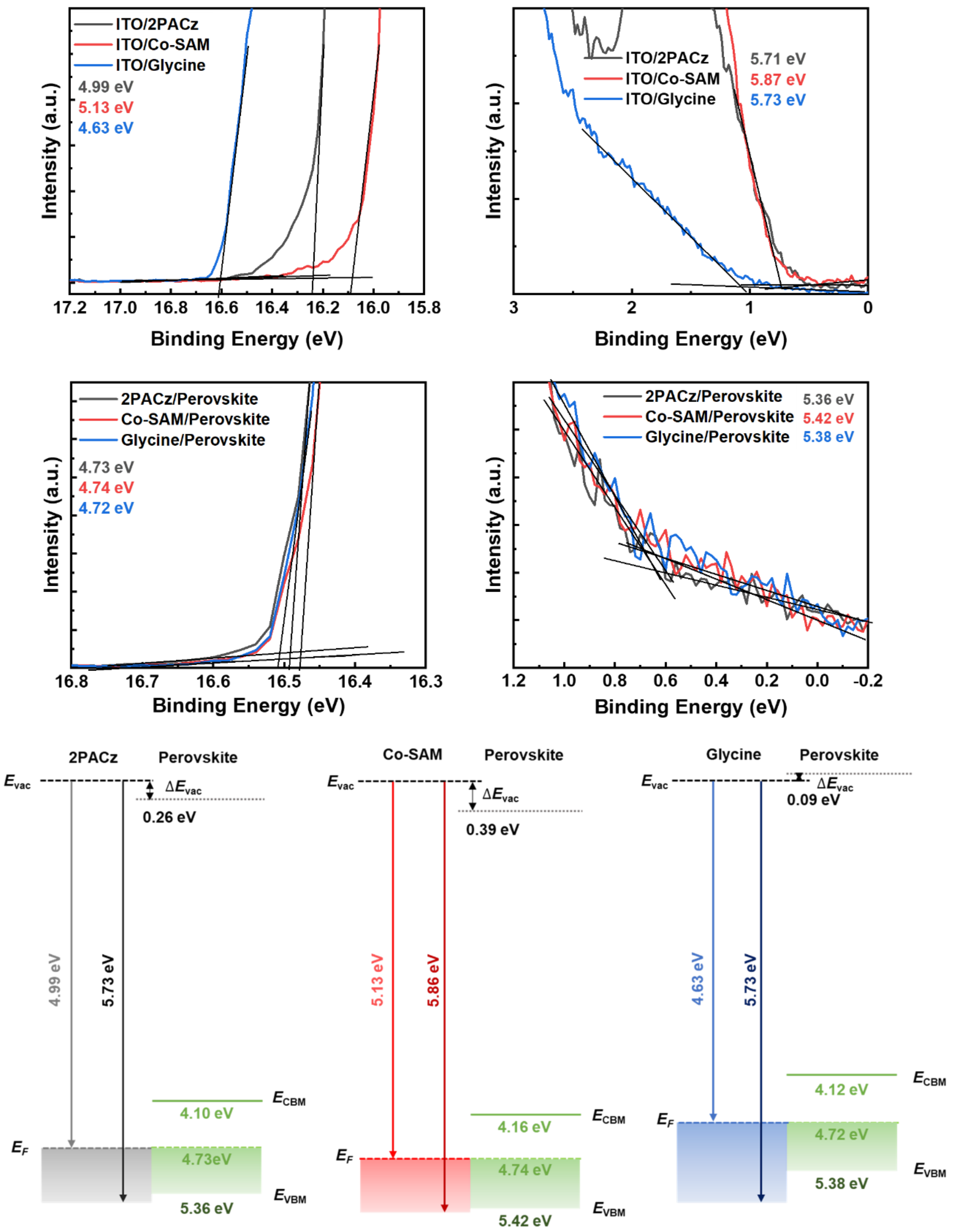

Figure S17. Cutoff region and valence band region of (a) ITO/SAM and (b) Sn-Pb perovskite films coated on ITO/SAM in UPS spectra. (c) The schematic energy-level diagrams for the ITO/SAM/perovskite.

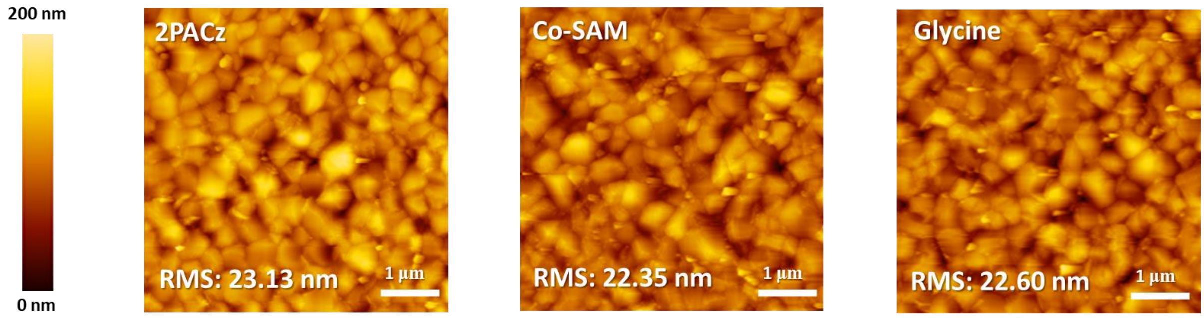

Figure S18. AFM images of perovskite films deposited on (a) ITO/2PACz, (b) ITO/Co-SAM, and (c) ITO/glycine.

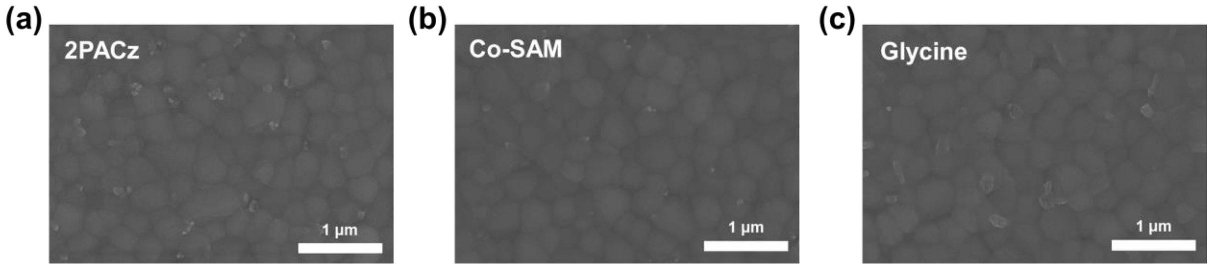

Figure S19. Top-view SEM images of perovskite films coated on (a) ITO/2PACz, (b) ITO/Co-SAM, and (c) ITO/glycine.

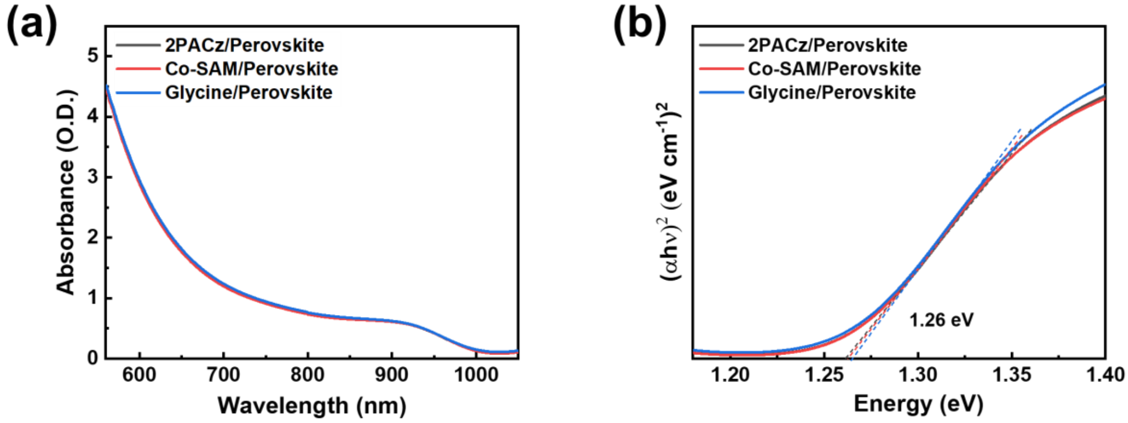

Figure S20. (a) UV-vis spectra and (b) Tauc plot of perovskite films deposited on ITO/2PACz (grey), ITO/Co-SAM (red), and ITO/glycine (blue).

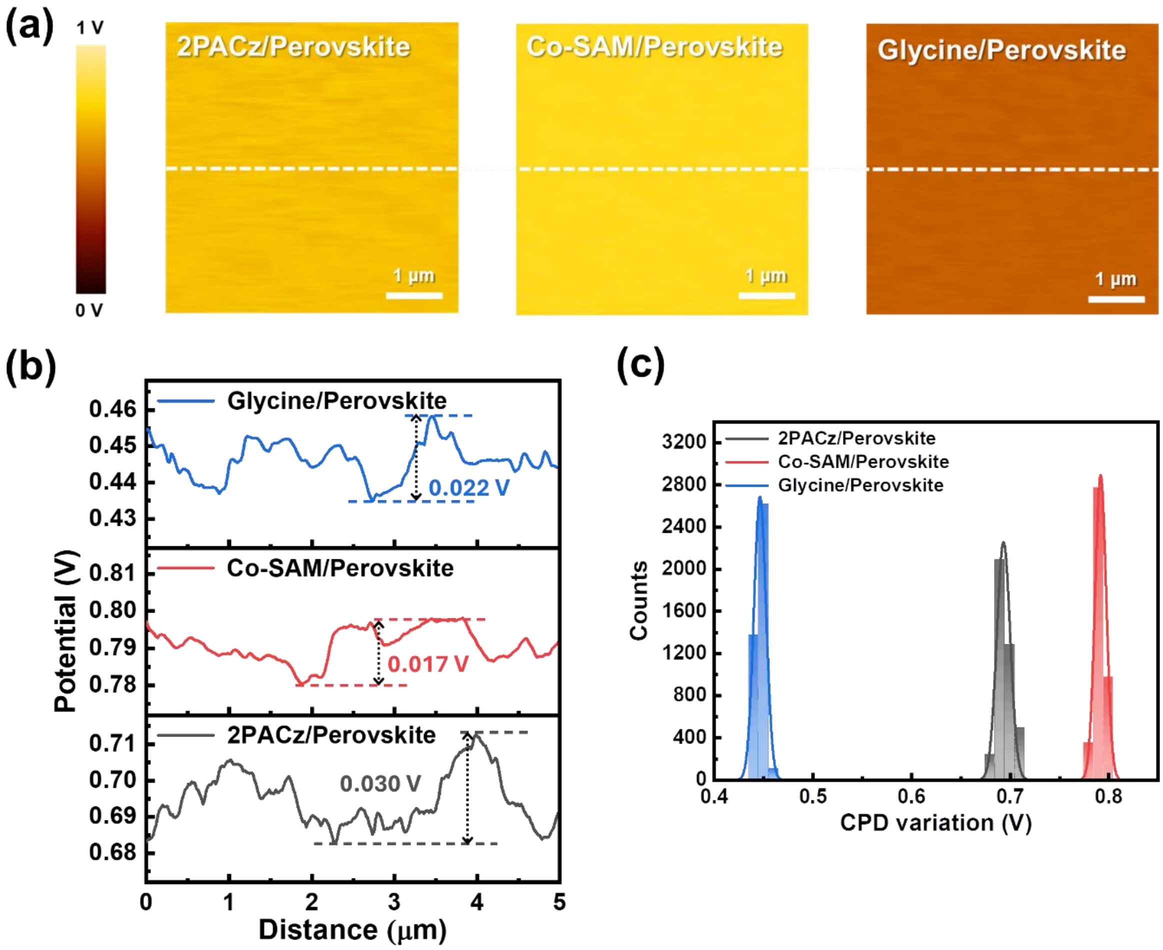

Figure S21. KPFM (a) surface potential images and the (b) corresponding line profiles for perovskite films coated on ITO/2PACz (grey), ITO/Co-SAM (red), ITO/glycine (blue). (c) Statistical variations of CPD from film surfaces obtained by KPFM.

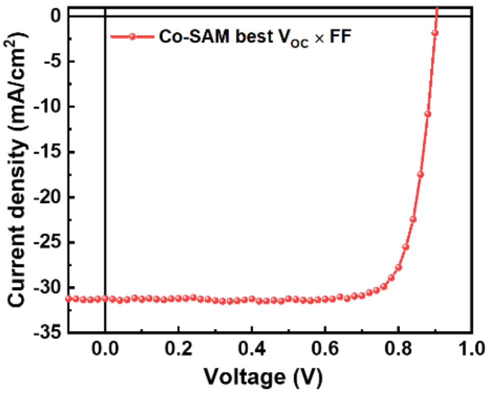

Figure S22. *J-V* curves of PSCs modified by Co-SAM, with derived the best $V_{OC} \times$ FF.

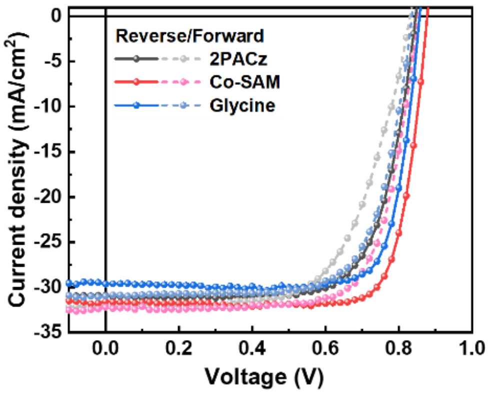

Figure S23. *J-V* curves for Sn-Pb PSCs with 2PACz, Co-SAM, and glycine, measured from open-circuit to short-circuit and back again.

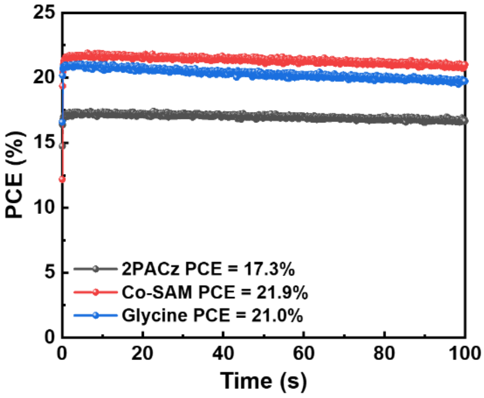

Figure S24. SPO of 2PACz, Co-SAM, and glycine-based Sn-Pb PSCs

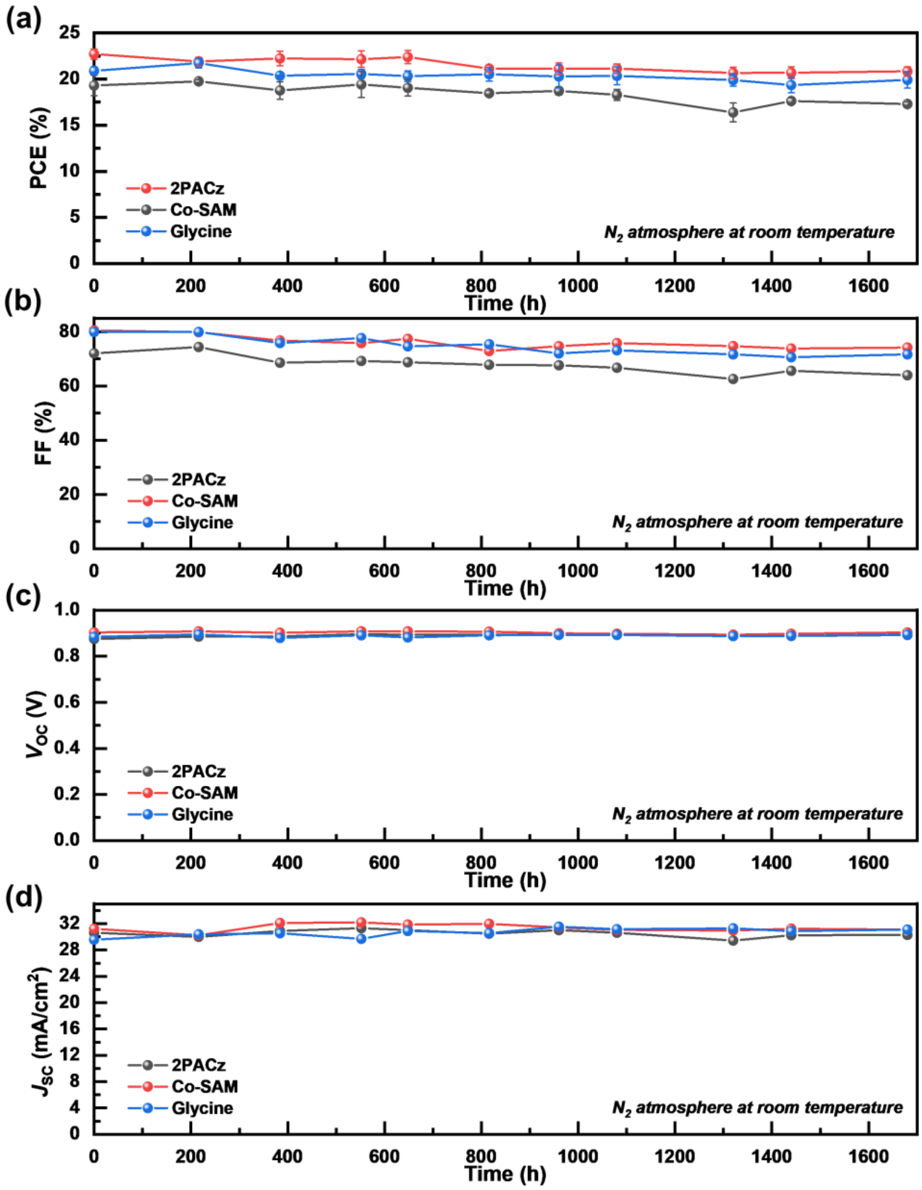

Figure S25. Long–term shelf-life stability of unencapsulated Sn-Pb PSCs modified by 2PACz, Co-SAM, and glycine at room temperature under dark conditions in a N_2 environment: (a) PCE, (b) FF, (c) V_{OC}, and (d) J_{SC}.

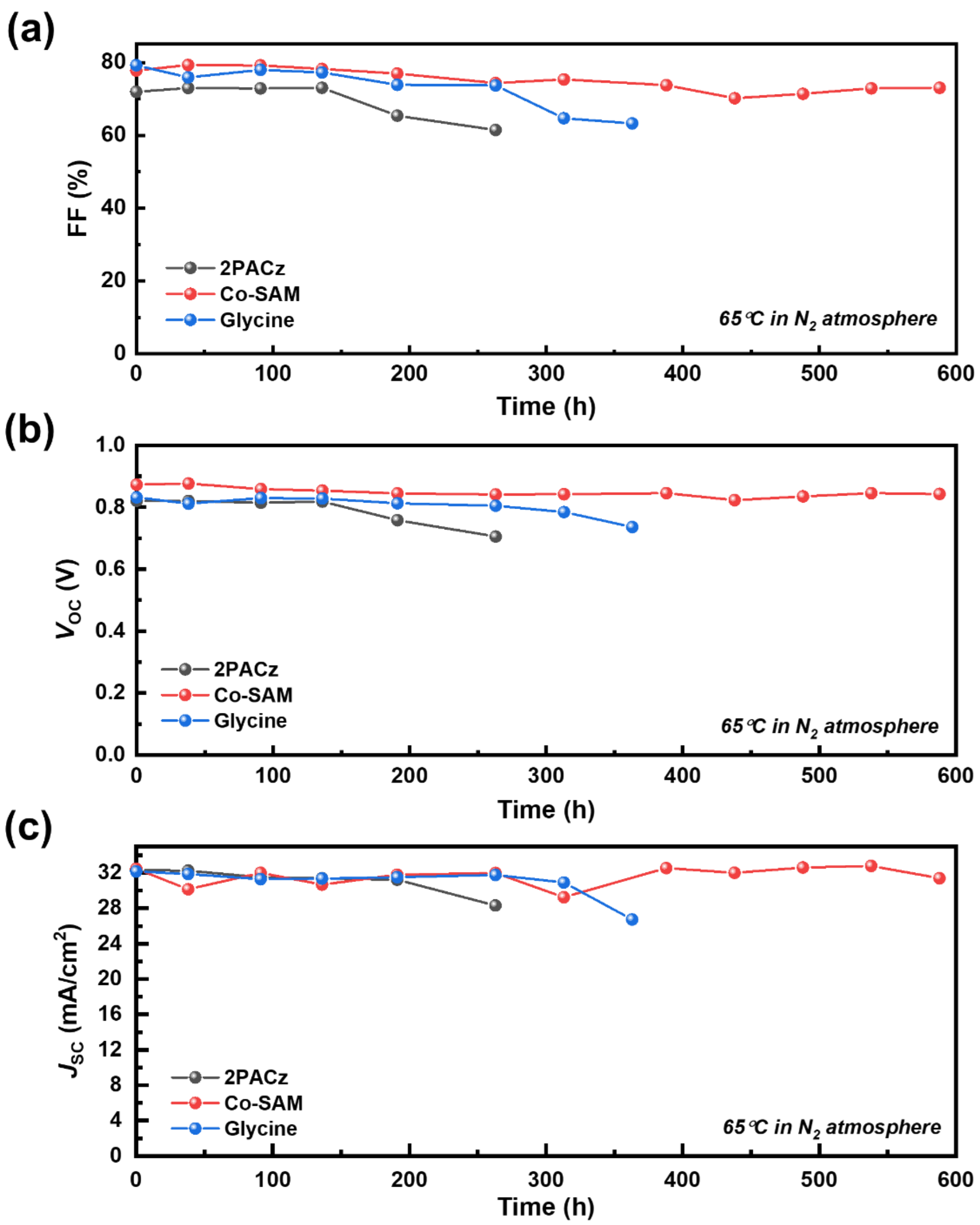

Figure S26. Thermal stability of unencapsulated Sn-Pb PSCs modified by 2PACz, Co-SAM, and glycine at 65 °C under dark conditions in a N_2 environment: (a) FF, (b) V_{OC}, and (c) J_{SC}.

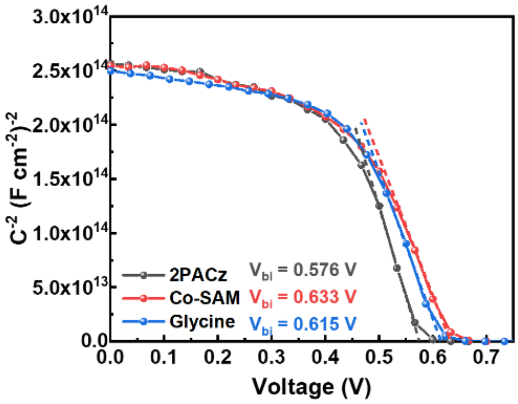

Figure S27. Mott–Schottky plots for Sn-Pb PSCs with 2PACz, Co-SAM, and glycine.

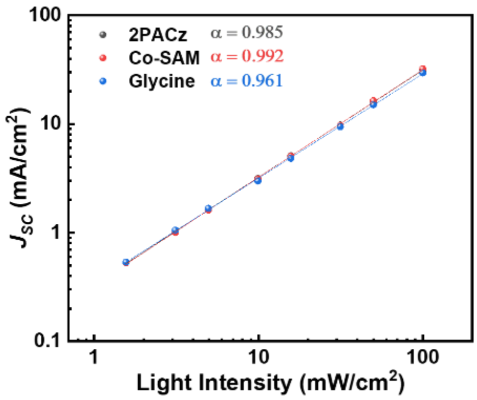

Figure S28. Light dependence of J_{SC} of Sn-Pb PSCs with 2PACz, Co-SAM, and glycine.

Peak	C-P	O-C=O
Binding Energy (eV)	286.93	288.48
Peak Area	921.31	771.38
Ratio	5.44	4.56

Table S1. Composition of Co-SAM coated on ITO substrates detected by C 1s XPS spectra. Peak positions and peak areas of C-P of 2PACz and O-C=O of glycine are demonstrated to calculate the ratio of 2PACz, and glycine molecules deposited.

	2PACz		Co-SAM		glycine	
Peak	Sn^{2+}	Sn^{4+}	Sn^{2+}	Sn^{4+}	Sn^{2+}	Sn^{4+}
Peak Area	50592.34	13273.27	42049.61	8948.78	43060.27	9316.19
Sn^{4+} Ratio	20.78%		17.55%		17.79%	

Table S2. Calculated ratio of Sn^{4+} corresponding to the Sn $3d_{5/2}$ XPS spectra of perovskite films coated on ITO/2PACz, ITO/Co-SAM, and ITO/glycine.

	2PACz		Co-SAM		glycine	
Peak	Sn^{2+}	Sn^{4+}	Sn^{2+}	Sn^{4+}	Sn^{2+}	Sn^{4+}
Peak Area	19366.95	13330.19	36885.67	10644.49	34903.69	12176.85
Sn^{4+} Ratio	40.77%		22.40%		25.86%	

Table S3. Calculated ratio of Sn^{4+} corresponding to the Sn $3d_{5/2}$ XPS spectra of perovskite films coated on ITO/2PACz, ITO/Co-SAM, and ITO/glycine after aging at 65 ℃ for 40 hours.

	J_{SC} [mA/cm^2]	$J_{SC\ cal.}$ [mA/cm^2]	V_{OC} [V]	FF [%]	PCE [%]
2PACz	32.61	31.37	0.857	76.63	21.43
Co-SAM	32.85	31.43	0.884	80.76	23.46
glycine	31.86	30.91	0.873	79.27	22.05

Table S4. Photovoltaic parameters of representative photovoltaic cells with 2PACz, Co-SAM, and glycine.

A-site composition	E_g [eV]	PCE [%]	J_{sc} [mA/cm^2]	V_{OC} [V]	FF [%]	Ref.
FACs	1.26	23.46	32.85	0.88	80.8	Present Work
FAMACs	1.25	23.17	32.99	0.91	77.2	[1]
FAMACs	1.27	22.15	31.59	0.87	81.1	[2]
FACs	1.24	19.51	32.1	0.81	75.0	[3]
FACs	1.24	22.41	30.78	0.88	82.7	[4]
FAMACs	1.25	22.67	31.98	0.89	79.3	[5]
FAMA	1.26	22.46	31.69	0.89	79.9	[6]
FACs	1.21	23.20	32.20	0.89	80.9	[7]
FAMA	1.24	18.59	30.66	0.79	76.6	[8]
FAMA	1.26	21.30	32.60	0.85	77.0	[9]
FAMA	1.25	22.20	30.62	0.92	79.0	[10]
FACs	1.22	22.60	32.64	0.87	79.6	[11]
FACs	1.21	21.20	31.50	0.86	78.1	[12]
FAMACs	1.26	22.30	31.20	0.88	81.0	[13]
FAMA	1.22	22.20	33.00	0.84	80.0	[14]
FACs	1.22	22.0	31.5	0.86	81.3	[15]
FAMACs	1.25	23.24	32.55	0.87	81.9	[16]
FACs	1.28	19.40	32.00	0.86	70.6	[17]
FAMA	1.25	22.16	31.42	0.88	79.8	[18]
FAMA	1.25	22.50	32.02	0.91	77.1	[19]
FAMA	1.25	23.80	33.00	0.87	82.6	[20]
FACs	1.21	23.70	33.80	0.88	80.0	[21]
FAMACs	1.25	21.65	31.82	0.85	79.8	[22]

Table S5. Device performance of reported Sn-Pb PSCs fabricated on ITO.

Scan direction		J_{SC} [mA/cm^2]	V_{OC} [V]	FF [%]	PCE [%]	HI [%]
2PACz	Reverse	30.89	0.846	72.02	18.83	10.09
	Forward	32.06	0.831	63.55	16.93	
Co-SAM	Reverse	31.41	0.880	80.00	22.12	9.14
	Forward	32.21	0.853	73.16	20.10	
glycine	Reverse	29.65	0.857	79.84	20.29	9.37
	Forward	30.96	0.836	71.02	18.38	

$$\boldsymbol{Hysteresis\ index\ (HI) = \frac{PCE_{Rev.} - PCE_{For.}}{PCE_{Rev.}}}$$

Table S6. Photovoltaic parameters of Sn-Pb PSCs with 2PACz, Co-SAM, and glycine, measured from open-circuit to short-circuit and back again and hysteresis index.

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(1) Zhang, W.; Yuan, H.; Li, X.; Guo, X.; Lu, C.; Liu, A.; Yang, H.; Xu, L.; Shi, X.; Fang, Z.; et al. Component Distribution Regulation in Sn-Pb Perovskite Solar Cells through Selective Molecular Interaction. *Advanced Materials* **2023**, *35* (39), 2303674. DOI: https://doi.org/10.1002/adma.202303674.

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